Sylius

Sylius is an open-source e-commerce framework that provides a modular and flexible structure for building customized online stores. Sylius is highly configurable and includes advanced features for catalog management, orders, payments, shipping, and other aspects of e-commerce. It is widely used by the developer community to create scalable and adaptable online stores. In this exercise, we will perform GUI testing on a newly created Sylius store (showroom/demo store) using the Cypress and Selenium frameworks.

GUI Testing Tools

Cypress is an open-source GUI test automation framework that enables writing and executing test scripts quickly and reliably. It stands out for its browser-based execution approach, allowing real-time interaction with interface elements. In addition, it provides an intuitive API for writing tests in JavaScript and supports several assertions to verify expected results.

Selenium is an open-source framework for GUI test automation in web applications. It allows interaction with GUI elements, simulates user actions, and automatically verifies results. Among its components, Selenium WebDriver is the most prominent, supporting several programming languages and enabling automation across different browsers. In this exercise, we will use Selenium with the JavaScript language and Mocha.

Target Application

Exercise Description

To complete the exercise, you will use the demo store provided by Sylius. The store has two parts: the storefront and the administrative panel.

In the storefront, users can perform actions related to purchasing products, such as searching for products, adding them to the shopping cart, checking out, etc.

In the administrative panel, we can manage the storefront by creating and editing products, categories, coupons, and many other features.

To simplify the use of the target application, a Docker image was created that can start all required services for the application (PHP, Nginx, and database), as well as install the latest version of the demo store, all within a single container. This image was uploaded and made available in a private registry.

The Dockerfile for this image can be found in this repository, along with the commands required to pull the image, run the container, and load sample data into the database (details in README2.md).

It is necessary to have Docker installed on your machine. We recommend using a Linux distribution such as Ubuntu.

If you have difficulty starting the application, seek assistance from José Neto (joserocha@copin.ufcg.edu.br).

Objective

Each student will be assigned (see spreadsheet) to test a specific section of the Sylius administrative panel (products, categories, coupons, etc.).

To do this, you must develop at least 10 test cases using Cypress and the same test cases using Selenium, totaling at least 20 test cases.

Each test case must contain at least 4 interactions with the application, meaning four actions performed on the website, such as clicking buttons, filling inputs, navigating between pages, etc.

In addition to the interactions, each test case must include at least one assertion to validate whether the expected result was achieved.

You are free to create tests across both parts of the store. For example, you may create a new product in the administration panel and then add that product to the cart in the storefront.

The objective is to develop a test suite for the assigned section, covering different functionalities and usage scenarios of that section.

To complete the exercise, you must start from this repository where Cypress and Selenium are already configured and extend the existing test scripts for both tools according to the assigned sections.

The scripts already contain the first test case as an example, meaning it is impossible to score zero. You must complete the scripts with at least 9 additional test cases for each framework.

Examples:

- `cypress/e2e/products.cy.js` for Cypress
- `selenium/e2e/products.js` for Selenium

Dependency to help implement tests in Cypress:

[Testing Library for Cypress]

Environment Setup (Windows)

Prerequisites

Install or have available on Windows:

- Docker Desktop (Linux containers mode)
- Git
- Node.js (LTS version, includes npm)
- VS Code (optional, but recommended)

Starting Sylius (application to be tested)

1.1 Open Docker Desktop

- Open Docker Desktop
- Wait until it shows “Docker Desktop is running”
- Make sure it is running in Linux containers mode

1.2 Sylius Project Structure

Navigate to the folder where your project is located.

1.4 Start Containers

Inside the Sylius folder:

```
docker compose up -d
docker compose ps
```

1.5 Install PHP Dependencies (Composer)

```
docker exec -it sylius-project-php-1 composer install
```

1.6 Compile Assets (Admin/Shop) with Node

Using local Node.js:

```
npm install
npm run build
```

Exercise Submission

The final deadline for completing the exercise is 03/04/2026.

What must be submitted

- This updated repository containing your implemented test cases and configurations
- Complete the evaluation form

Useful Links

- Cypress Documentation
- Selenium Documentation
- Examples created during class